

Data Modeling in Power BI

Beginner-friendly student notes

1. What is Data Modeling?

Simple explanation

"Data modeling means organizing your tables and connecting them correctly so Power BI understands how they relate to each other."

When you load data from Excel, SQL, or any other source into Power BI, you usually get several separate tables — for example, Sales, Products, Customers, and Dates. Data modeling is the process of arranging these tables and building relationships between them so that a report can pull the right numbers from the right tables automatically.

Why it matters

- Wrong or missing relationships give wrong totals in visuals.
 - A clean model makes DAX formulas simpler to write.
 - A well-modeled report is faster and easier to maintain.
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2. Fact Tables vs Dimension Tables

Every Power BI model is usually built from two kinds of tables:

Fact Table

Purpose

Stores the numbers/events you want to measure — things that happen repeatedly.

Example

A Sales table with columns like Order_ID, Product_ID, Customer_ID, Date, Quantity, Amount.

Dimension Table

Purpose

Stores descriptive details used to filter, group, or label the facts.

Example

A Products table with Product_ID, Product_Name, Category. A Customers table with Customer_ID, Customer_Name, City.

Rule of thumb

"Fact tables hold numbers you add up. Dimension tables hold labels you filter or group by."

	Fact Table	Dimension Table
Contains	Transactions / measurements	Descriptive attributes
Example columns	Amount, Quantity, Order_Date	Product_Name, City, Category
Size	Usually large, grows over time	Usually smaller, changes rarely
Example table	Sales	Products, Customers, Dates

3. Relationships

Purpose

A relationship links two tables using a common column, so filtering one table automatically filters the other.

Simple explanation

| "A relationship is like a connecting wire between two tables, plugged in at the matching column."

Example

Sales.Product_ID connects to Products.Product_ID. If a user filters a report by Category (from Products), Power BI uses this relationship to filter the Sales table too.

How it's created

In Power BI: Model view → drag Product_ID from Sales onto Product_ID in Products. Power BI detects the key columns and draws the connecting line automatically.

4. Cardinality (Type of Relationship)

Cardinality describes how many matching rows exist on each side of a relationship.

One-to-Many (most common)

Simple explanation

| "One row on one side can match many rows on the other side."

One row in Products (Product_ID = 1) can appear many times in Sales, since the same product is sold many times.

One-to-One

Simple explanation

| "Each row on one side matches exactly one row on the other side."

An Employees table and an EmployeeLogins table, where each employee has exactly one login record.

Many-to-Many

Simple explanation

| "Many rows on one side can match many rows on the other side."

A Students table and a Courses table, where each student takes many courses and each course has many students — usually needs a bridge table in between.

5. Cross Filter Direction

Purpose

Controls which direction a filter travels across a relationship.

Single direction

Simple explanation

| "Filtering flows one way — from the '1' side (dimension) to the 'many' side (fact)."

Filtering Products by Category filters Sales. This is the default and recommended setting for most models.

Both directions

Simple explanation

| "Filtering flows both ways between the two tables."

Used only in special cases, since it can slow down the model and sometimes causes ambiguous results. Use single direction unless there is a clear reason not to.

6. Star Schema vs Snowflake Schema

Star Schema

Simple explanation

| "One central fact table, surrounded directly by dimension tables — shaped like a star."

Sales (fact) connects directly to Products, Customers, and Dates (dimensions). This is the recommended structure for Power BI — simple, fast, and easy to understand.

Snowflake Schema

Simple explanation

| "Dimension tables are further split into smaller related tables, creating extra layers of relationships."

Products connects to Categories, and Categories connects to Departments, instead of keeping everything in one Products table. This adds complexity, so it's generally avoided unless there's a strong reason for it.

	Star Schema	Snowflake Schema
Structure	Fact table directly linked to dimensions	Dimensions split into sub-dimensions
Complexity	Simple	More complex
Performance	Generally faster	Can be slower (more joins)
Recommended for Power BI	Yes	Only when necessary

7. Primary Key & Foreign Key

Primary Key

Purpose

A column that uniquely identifies each row in a table.

Example

Product_ID in the Products table — no two products share the same Product_ID.

Foreign Key

Purpose

A column in one table that refers to the primary key of another table, used to build the relationship.

Example

Product_ID in the Sales table refers back to Product_ID in Products.

8. Best Practices for Data Modeling

- Prefer a star schema over a snowflake schema when possible.
- Keep relationships as one-to-many with single-direction filtering wherever possible.
- Hide foreign key columns from report view once relationships are built, to avoid confusing users.
- Use a separate Dates table for all date-based analysis, connected to every date column that needs it.
- Give tables and columns clear, business-friendly names.
- Avoid unnecessary columns — only bring in what the report actually needs.

Quick Revision

Term	Main Idea
Data Modeling	Organizing and connecting tables correctly
Fact Table	Stores numbers/events (e.g., Sales)
Dimension Table	Stores descriptive labels (e.g., Products, Customers)
Relationship	Connects two tables using a common column
Cardinality	How many rows match on each side (1:1, 1:Many, Many:Many)
Cross Filter Direction	Which way a filter travels across a relationship
Star Schema	Fact table directly linked to dimensions — recommended
Snowflake Schema	Dimensions split into further sub-tables — avoid unless needed
Primary Key	Uniquely identifies each row in a table
Foreign Key	Refers to a primary key in another table

One-line concept to remember:

Data modeling = organize tables (fact vs dimension) + connect them (relationships) in the simplest shape possible (star schema).